Bachelor Project Applied Computer Science
(Academic year 2025—2026)

Federated Learning

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Contents

1 Scope & Vision

1.1 Background

What is the background of the company for which you carried out this project? Not applicable if the project was not conducted for a company.

Why would the company want this to be researched or developed? If not for a company: what is the motivation behind the bachelor project?

1.2 Problem statement

What is the problem statement?

1.3 Intention

What is the company's intention? If not for a company: how will the product be used in the long term (independently of a company)?

1.4 Scope within the bachelor project (MVP – need to have)

What is the scope of the project within this bachelor project?

1.5 Ultimate scope (want to have)

What is the ultimate scope?

2 Software Requirements Specification

2.1 Target group

Who are the target groups of the tool to be developed?

What aspects must you consider for these target groups?

What are the needs of these target groups?

2.2 Method

What form of requirements gathering did you use?

2.3 Requirements

What are the functional requirements?

Which features/user stories did you design?

What are the non-functional requirements?

Which features/user stories belong to the MVP?

3 Preliminary Research

3.1 Related projects

Are there any related problem statements?

3.2 Related solutions

Are there already similar digital solutions or tools that address this?

Can existing digital solutions be (partially) reused?

3.3 Required technology

What technology is required to develop the solution or your digital product?

Which programming languages did you use?

Which programming languages did you test but ultimately not use in the final development?

3.4 Comparative study

Did you conduct a comparative study of suitable technologies?

Is there a justification for the technologies that were selected?

3.5 Final selected technology / project tools

Which project tools did you use?

Which development methodologies did you apply?

3.6 Choice reflection

4 Reflection

4.1 General Reflection

4.1.1 Achieved objectives

Were the project objectives achieved?

4.1.2 Test framework / technique

Which test framework or technique was chosen and why?

Which parts of the project were or were not tested, and why?

4.1.3 Security

Was security considered in the project and how was it addressed?

Where do you see possible improvements in the project's security?

4.1.4 Improvements

Where do you see improvements in general?

4.1.5 Added value of the project

What do you consider the greatest added value of the project?

4.1.6 Future opportunities

What are the next steps or future opportunities for the project?

4.1.7 Next steps

What are the next steps or future opportunities for the project?

4.2 Personal Reflection

If you were to do the project again, what would you approach differently?

What impact do you think the research or developed tool has on people (ethics/inclusion)?

What impact do you think the research or developed tool has on society (economy/policy/education)?

What impact do you think the research or developed tool has on the climate (energy/sustainability)?

What will stay with you most about the project?

4.3 Feedback

What did you think overall about the integrated bachelor project and the distribution with the traineeship?

Do you have suggestions for improvement for the integrated bachelor project?

5 Appendices

Meeting minutes van vergaderingen

The backlog

The story mapping (with different releases)

Example mappings (if available or in another format)

Gherkin scenarios (if available)

Login credentials

Installation manual

User manual

5.1 Meeting minutes

5.2 Backlog

5.3 Story mapping

References

[Gre93] George D. Greenwade. The Comprehensive Tex Archive Network (CTAN). *TUGBoat*, 14(3):342–351, 1993.